

Seamless Switching Technology of Pixel Clock on AMOLED Notebook Display

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Abstract

Notebooks have continuously adopted innovative technologies to meet two key requirements: improving visual performance and extending battery life. When the system frame rates decrease, the pixel clock and embedded DisplayPort (eDP) transmission rates are adjusted. LCD Seamless Panel Refresh Switching (SPRS) is widely implemented. However, OLED differs from LCD by integrating light-emitting units and pixel control units. Direct application of LCD SPRS causes critical issues, including screen blackouts, flicker, and tearing.

This paper presents a timing regeneration solution for OLED notebook. It has implemented the industry's first eye-invisible pixel clock switching on OLED notebook. The solution reduces logical power by ~102 mW (a power drop of 5.3%). Since eDP operates between the source-end and the sink-end, the benefit of this solution is approximately double (it reduces ~200 mW).

Author Keywords

Notebook (NB); Pixel clock; eDP; Seamless switching; Timing Controller (TCON); Graphics Processing Unit (GPU).

1. Introduction

In OLED NB, the key display ICs are GPU, TCON, and source driver. Typically, reducing the panel refresh rate can achieve lower power by reducing eDP bandwidth, decreasing the TCON internal operating frequency, and enabling the wake-up function between TCON and source driver. As shown in Figure 1.

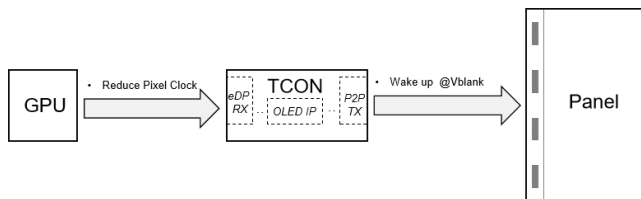


Figure 1. Diagram of OLED NB module.

Currently, OLED NB achieves refresh rate switching via the long_V solution. This technology maintains a fixed V_Active duration, ensuring a consistent pixel charging time per row (1H = V_Active time/Gate lines). The IPs compensation parameters (Digital Gamma, De-Mura, De-burn-in, Over Driving, etc.) remain consistent across different refresh. As shown in Figure 2.

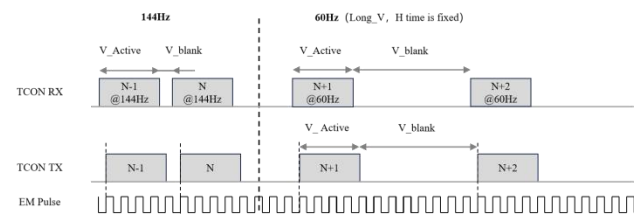


Figure 2. The timing of current long_V of OLED NB.

However, OLED NB lacks compatibility with the long_H solution commonly employed in LCD NB. This is due to varying pixel charging times during frequency switching. Additionally, numerous compensation IP parameters cannot be shared, leading to image quality deterioration. This also makes precise control of EM Duty position and width nearly impossible, causing screen flicker and impairing user experience. As shown in Figure 3.

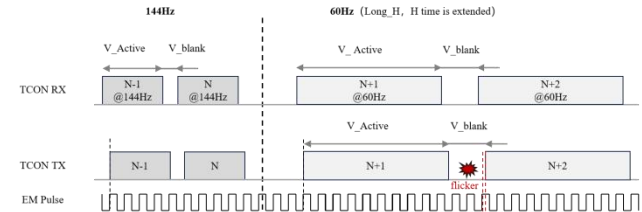


Figure 3. Challenges of traditional Long_H in OLED NB.

To address this challenge, the authors present a solution that combines the long_V solution from current OLED NB with the long_H solution from LCD NB. The TCON RX processes the low-rate pixel clock transmitted by the GPU (eDP signal rate downshifting) and temporarily stores the corresponding image data in an additional available SPRS buffer. Concurrently, to maintain user visual performance, the TCON and Panel sustain the original pixel clock, thereby enabling maximal power reduction while preserving display quality. As shown in Figure 4.

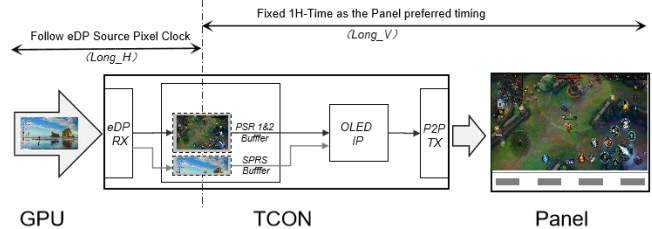


Figure 4. The solution combining long_V with long_H.

The characteristics and practical performance of these three low-power solutions are summarized in Table 1. The authors' proposal achieves a better balance between lower power and NB user acceptable display quality.

Table 1. Comparison of three low-power solutions

Solution	OLED Low Power @Low Refresh Rate			Display Quality
	Reduce eDP Bandwidth	Reduce TCON Internal Freq	Wake-up @ Vblank	
Long_V	No Support	No Support	★ ★ ★	OK
Long_H	★ ★	★ ★ ★ ★	☆	NG
Combine H&V	★ ★	No Support	★ ★ ★	OK

2. Application and Results

Throughout the process, the key mechanism involves TCON capability to dynamically regulate interactions with upstream and downstream ICs, facilitating seamless link training of eDP interface while maintaining orderly storage and accurate display of image content. The solution combining long_H&V is primarily composed of three stages: 144 Hz pixel clock, static image display, and 60Hz pixel clock. The EM Pulse maintains a constant frequency throughout the whole operation (144Hz@5 EM pulse/60Hz@12 EM pulse). The role and behavior of each stage are described in detail. As shown in Figure 5.

At the first stage named as 144Hz pixel clock, the full-link training signal maintains a stable low level, indicating successful link initialization. The GPU, TCON, and OLED panel operate at 144Hz frame rate and refresh rate. The eDP rate operates at HBR3 (8.1Gbps). Concurrently, TCON enters PSR1 (Panel Self-Refresh) mode for storing the frame_N into PSR frame buffer. The RX and TX of TCON sequentially process the frames (N-1) and frames_N, ensuring real-time display of dynamic content.

At the second stage named as static image display, TCON sustains refreshing at the original 144Hz refresh rate by utilizing frame_N of data from the PSR frame buffer. When the full-link training signal is enabled, the GPU transmits the training patterns synchronized to the 60Hz pixel clock to the TCON. Upon successful link establishment, the full-link training signal is disabled. After this juncture, the system is required to switch to re-sync mode and disable the PSR1 function.

At the final stage named as 60Hz pixel clock, the GPU start transmitting 60Hz content to TCON. The pixel clock reduces from 851.38 MHz to 354.78 MHz, while the eDP rate decreases from HBR3 to HBR. This achieves substantial power savings. The TCON RX operates at a reduced reception rate, while the TCON TX maintains a higher transmission rate. To prevent screen tearing and flicker, the leading portion (about 1050 line) of the image requires temporary buffering in the TCON SPRS RAM buffer. This necessitates coordination among the TCON, GPU, and eDP protocol to leverage idle frame buffers effectively.

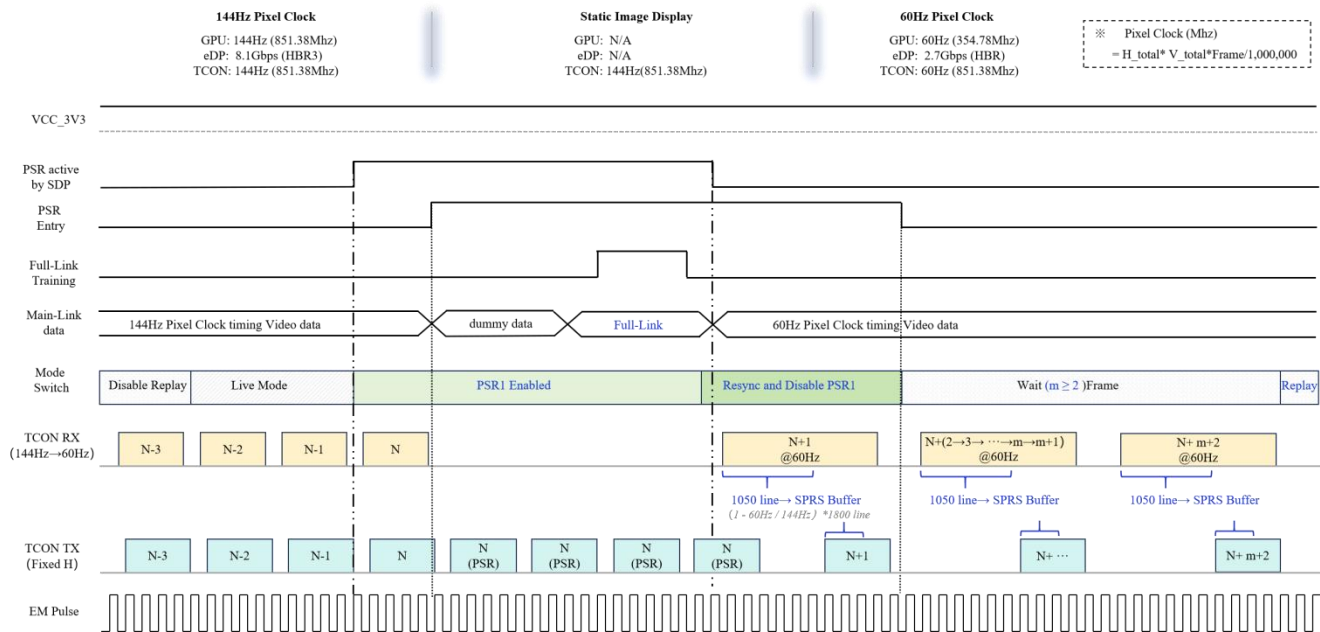


Figure 5. The timing of the solution combine H&V (2880RGB*1800).

The key advantage of this solution is its ability to achieve low power without resetting VCC_3V3. This requires precise coordination between the internal and external clock frequency.

It is essential to explain the process of reading and writing display data in the TCON's internal SRAM during frame rate switching.

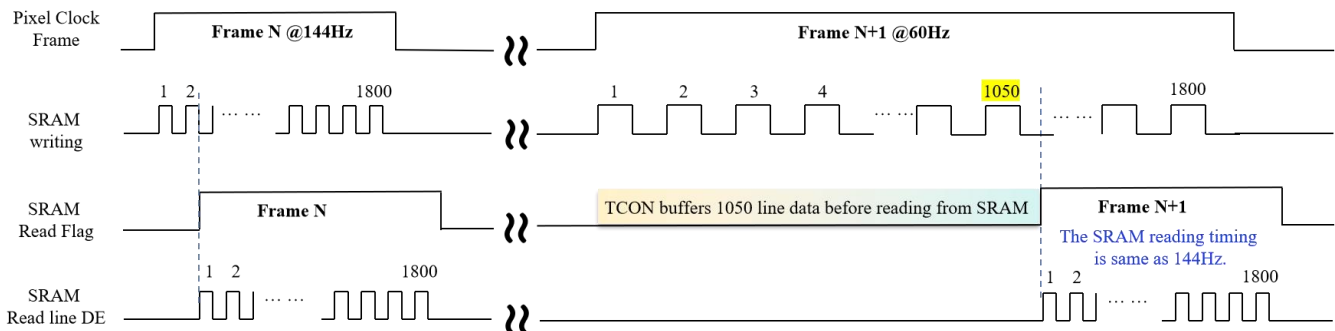


Figure 6. The process in TCON' SRAM of the solution combine H&V (2880RGB*1800).

The authors measured the power consumption of the OLED NB product (14 - inch, 2880 RGB*1800@60~144 Hz, 3 pcs) using a digital multi-meter (Keysight 34461A). As shown in Figure 6.

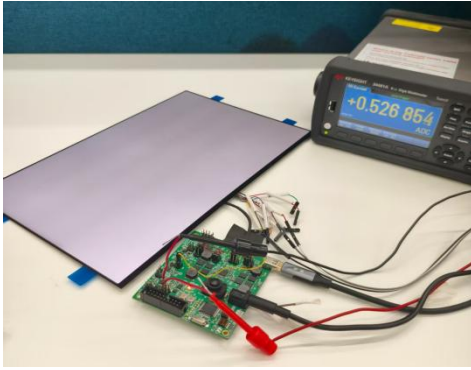


Figure 7. The OLED NB module under test.

The power consumption under multiple test pattern at 144Hz is taken as the corresponding baseline (100%). Compared to the long_V, the combined H&V can achieve a power consumption

benefit of approximately 102 mW at 60 Hz. As shown in Table 2,

Table 2. Power consumption of three low-power solutions

Power /W	144Hz	60Hz		
	Baseline	Long_V	Long_H	Combine_H&V
Desktop_Win11	2.07 (100%*A)	1.46 (70.84%*A)	1.13 (54.70%*A)	1.36 (65.72%*A)
White_255	1.95 (100%*B)	1.39 (71.42%*B)	1.06 (54.58%*B)	1.29 (66.09%*B)
Chess_8*8	1.92 (100%*C)	1.36 (71.00%*C)	1.03 (53.68%*C)	1.26 (65.53%*C)

For the White_255 pattern example, long_V is 71.42% and combined long_H&V is 66.09%. The solution of combined long_H&V reduces logical power by ~102 mW (a total logic power drop of 5.3%), as shown in Figure 6. Notably, long_H exhibits the lowest power-saving capability. However, the brightness delta of pre-/post-pixel clock switching is over 30 nits. Consequently, this solution of long_H is not viable for practical applications, and the data is for reference only.

Power Ratio Comparison of Three Low-Power Solutions Under Multiple Test Patterns

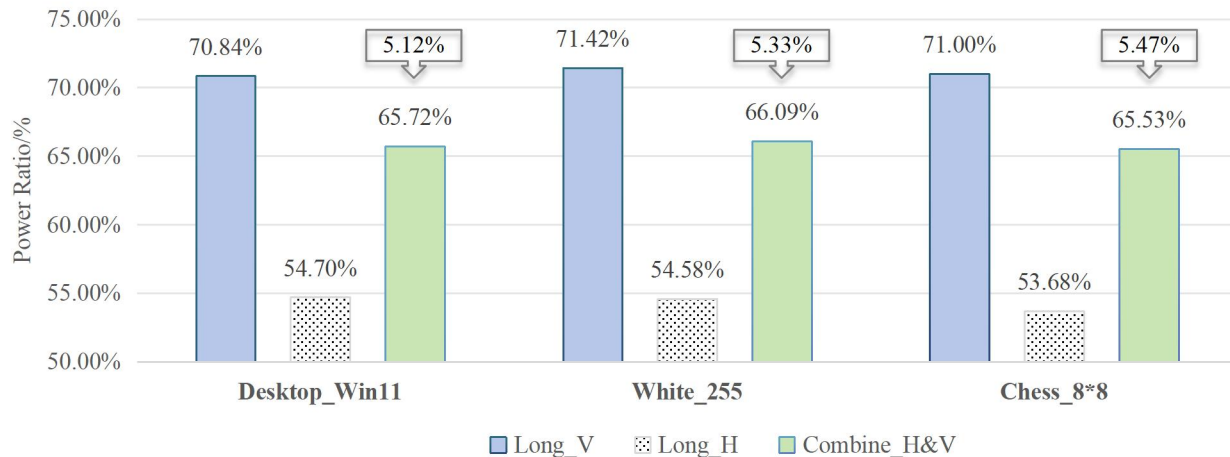


Figure 8. Power ratio comparison of three low-power solutions under multiple test patterns

3. Conclusion and Impact

To meet the dual requirements of improving visual performance and extending battery life of NB products, the authors developed the combine_H&V solution by analyzing LCD SRPS technology (long_H) and incorporating OLED unique driving characteristics. This solution achieves ~102mW power savings compared to the original OLED long_V solution while maintaining display quality (delta luminance < 1%). The value of 102mW is equivalent to 5.3% of the MDL total power consumption. Since eDP power reduction benefits not only the Sink Device (TCON) but also the Source Device (GPU), the solution is estimated to achieve approximately 200mW in overall.

Because of an additional SPRS buffer, this solution may increase the cost of TCON IC. Given its close association with the PSR behavioral logic of the NB system and application scenarios, this solution requires the collaboration among upstream and downstream (e.g., CPU, GPU, TCON IC, and panel suppliers) to continuously refine verification conditions and optimize the

solution. The combine_H&V solution will be more competitive if it can be implemented by leveraging the existing PSR frame buffer through well - defined application scenarios, strict timing control, and flexible IC hardware design. If you have any suggestions for improvement, please contact the author by email.

The eDP standard protocol is widely adopted in NB products. It features an embedded clock and multiple transmission speed rate options (e.g., HBR, HBR2, HBR3). However, most tablets and smartphones utilize the MIPI (D-Phy) protocol, which has a unique low-power mode (LP) and Source Device (GPU&CPU) synchronization scheme. MIPI's clock signal is transmitted in parallel with data signals through a dedicated differential pair. At low refresh rates, MIPI's clock channel suspends high-speed toggling and switches to non-continuous toggling, while the data channel remains active but switches to low-power state. Therefore, the low-power solution of combine_H&V for OLED NB is unlikely to be extended to tablets or smartphones in the near future.

4. Acknowledgements

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5. References

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